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15.2.2 Rules for function overloading

1. The overloaded function must differ in the number of its arguments or data types
2. The return type of overloaded functions are not considered for overloading same data type
3. The default arguments of overloaded functions are not considered as part of the parameter list in function overloading.

Illustration 15.2 C++ Program to demonstrate function overloading

```
#include <iostream>
using namespace std;
long add(long, long);
long add(long,long,long);
float add(float, float);
int main()
{
 long a, b, c,d;
 float e, f, g;
 cout << "Enter three integers\n";
cin >> a >> b>>c;
d=add(a,b,c); //number of arguments different  but  same data type
cout << "Sum of 3 integers: " << d << endl;
cout << "Enter two integers\n";
cin >> a >> b;
c = add(a, b); //two arguments data type same with above function call
                                        and different with below function call
cout << "Sum of 2 integers: " << c << endl;
cout << "Enter two floating point numbers\n";
 cin >> e >> f;
 g = add(e, f); //two arguments  similar to the above function call but data
                                                        type different
cout << "Sum of floats: " << g << endl;
}
long add(long c, long g)
{
 long sum;
 sum = c + g;
 return sum;
}
```